



DEEPAK FERTILISERS  
AND PETROCHEMICALS  
CORPORATION LIMITED

## MSDS - ISO PROPYL ALCOHOL

### SECTION 1 - CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

|                         |                                                               |
|-------------------------|---------------------------------------------------------------|
| Chemical Name           | : Iso Propyl Alcohol                                          |
| Chemical Formula        | : (CH <sub>3</sub> ) <sub>2</sub> CHOH                        |
| CAS Number              | : 67 – 63 – 0                                                 |
| Synonyms                | : 2 – Propanol, Iso Propanol, Sec – Propyl Alcohol            |
| General Use             | : As solvent                                                  |
| Manufacturer's Name     | : Deepak Fertilisers and Petrochemicals Corprn. Ltd.          |
| Address :               | : Plot K-1, MIDC Indl Area, Taloja A.V., Dist: Raigad, 410208 |
| Telephone no. for info. | : +91 - 022-50684000                                          |

### SECTION 2 - COMPOSITION / INFORMATION ON INGREDIENTS

|                      |                                 |
|----------------------|---------------------------------|
| Composition          | : Iso Propyl Alcohol 99%        |
| Hazardous components | : Iso Propyl Alcohol            |
| ACGIH TLV            | : 200 ppm (TWA), 400 ppm (STEL) |

### SECTION 3 - HAZARDS IDENTIFICATION

|                                        |                                                                                                                                                                                                                      |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary Entry Routes                   | : Inhalation, skin, eyes and ingestion                                                                                                                                                                               |
| Acute Effects                          | : Inhalation of vapours irritates the respiratory tract. Exposure to high concentration has a narcotic effect producing symptoms of dizziness, drowsiness, headache, staggering, unconsciousness and possibly death. |
| Carcinogenicity                        | : Not classified as human carcinogen                                                                                                                                                                                 |
| Chronic Effects                        | : Chronic exposure may cause skin effects.                                                                                                                                                                           |
| Aggravation of Pre-existing conditions | : Persons with pre-existing skin disorders or impaired liver, kidney or pulmonary function may be more susceptible to the effects of this chemical                                                                   |
| NFPA rating                            | : Health – 1; Fire – 3; Reactivity - 0                                                                                                                                                                               |

### SECTION 4 - FIRST AID MESURES

|            |                                                                                                                                                               |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Eyes       | : Immediately flush eyes with plenty of water for atleast 15 minutes<br>; Lifting lower & upper eyes lids occasionally,<br>Get medical attention immediately. |
| Skin       | : Immediately flush skin with plenty of water for atleast 15 minutes<br>; Call a physician, if irritation develops.                                           |
| Inhalation | : Remove affected person to fresh air.;If not breathing, give artificial respiration.<br>If breathing is difficult, give oxygen.<br>Get medical attention.    |
| Ingestion  | : Give large amount of water to drink.<br>Never give anything by mouth to an unconscious person.<br>Get medical attention.                                    |



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### SECTION 5 - FIRE FIGHTING MEASURES

|                               |                                                                                                                                                                                                                                                                          |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Flash point                   | : 12 oC (54o F)                                                                                                                                                                                                                                                          |
| Flash point method            | : Closed Cup                                                                                                                                                                                                                                                             |
| Auto ignition temp.           | : 399 oC (750o F)                                                                                                                                                                                                                                                        |
| LEL                           | : 2%                                                                                                                                                                                                                                                                     |
| UEL                           | : 12.5%                                                                                                                                                                                                                                                                  |
| Flammability classifn.        | : Flammable                                                                                                                                                                                                                                                              |
| Extinguishing media           | : Water spray, Dry chemical, Alcohol Foam or CO2<br>Water spray may be used to keep fire expose containers cool, dilute spills to non-flammable mixtures and to protect personnel attempting to stop leak and disperse vapours.                                          |
| Hazardous combustion products | : Above flash point, vapour air mixtures are explosive within flammable limits as mentioned above. Contact with strong oxidizers, may cause fire or explosion. Vapours can flow along surfaces to distant ignition source and flash back. Sensitive to static discharge. |
| Hazardous combustion products | : CO2 and Carbon Monoxide may be formed when heated to decomposition.                                                                                                                                                                                                    |
| Fire fighting instructions    | : In the event of fire; Wear full protective clothing and self contained breathing apparatus with full face piece.                                                                                                                                                       |

### SECTION 6 - ACCIDENTAL RELEASE MEASURES

|              |                                                                                                     |
|--------------|-----------------------------------------------------------------------------------------------------|
| Small Spills | : Shut off leaks without risk. Absorb on sand or earth.                                             |
| Containment  | : Prevents spillage from entering drains or water sources. Contain and recover liquid when possible |
| Clean up     | : Wash the area with water preventing run-off from entering drains.                                 |

### SECTION 7 - HANDLING AND STORAGE

|                      |                                                                                                                                                                                                             |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Handling Precautions | : Protect against physical damage; Containers should be bonded and grounded. Use non-sparking tools & equipment. Avoid breathing vapours. ; Avoid contact with skin & eyes. Wash thoroughly after handling. |
| Storage Requirements | : Store in a cool, dry, well ventilated area away from heat source. Storage and usage area should be clarified as "No smoking Area".                                                                        |

### SECTION 8 - EXPOSURE CONTROLS / PERSONAL PROTECTION

|                        |                                                                                                 |
|------------------------|-------------------------------------------------------------------------------------------------|
| Engineering Controls   | : Provide proper ventilation so as to maintain environment below air borne exposure limit.      |
| Respiratory Protection | : If exposure limit is exceeded, use respiratory protection                                     |
| Protective Clothing /  | : Wear impervious protective clothing including boots, gloves, apron, coveralls while handling. |



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### SECTION 9 - PHYSICAL AND CHEMICAL PROPERTIES

|                    |                                          |
|--------------------|------------------------------------------|
| Physical State     | : Liquid                                 |
| Appearance & Odour | : Colourless liquid with alcoholic odour |
| Vapor Pressure     | : 44 mmHg @ 25 oC                        |
| Specific Gravity   | : 0.79 @ 20 oC                           |
| Water Solubility   | : Miscible in water                      |
| Boiling Point      | : 82 oC                                  |
| Freezing Point     | : (-)88.4 oC                             |
| Vapor Density      | : 2.1                                    |

### SECTION 10 - STABILITY AND REACTIVITY

|                                 |                                                                                                                                                                                                                                                                  |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stability                       | : Stable under ordinary conditions of Use and Storage. Heat and sunlight can contribute to instability.                                                                                                                                                          |
| Chemical incompatibilities      | : Oxidizing agents, acetaldehyde, acids, chlorine, ethylene oxide, hydrogen-palladium combination, hydrogen peroxide-sulfuric acid combination, potassium t-butoxide, hypochlorous acid, iso cyanides, nitroform, phosgene, aluminum, oleum and perchloric acid. |
| Conditions to Avoid             | : Heat, flames, ignition sources and incompatibles.                                                                                                                                                                                                              |
| Hazardous Decomposition product | : CO <sub>2</sub> and Carbon Monoxide may be formed when heated to decomposition.                                                                                                                                                                                |

### SECTION 11 - TOXICOLOGICAL INFORMATION

|                  |                                                                                                                                                                                                                      |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TLV as per ACGIH | : 200 ppm (TWA), 400 ppm (STEL)                                                                                                                                                                                      |
| Acute-Inhalation | : Inhalation of vapours irritates the respiratory tract. Exposure to high concentration has a narcotic effect producing symptoms of dizziness, drowsiness, headache, staggering, unconsciousness and possibly death. |

### SECTION 12 - ECOLOGICAL INFORMATION

|                        |                                                                                                                                                                                                                     |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Environmental toxicity | : Prevent spillage from entering drains or water sources. After spills, wash area with water preventing run-off from entering drains. When released into water, this material may bio-degrade to a moderate extent. |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### SECTION 13 - DISPOSAL CONSIDERATIONS

|          |                                                                                      |
|----------|--------------------------------------------------------------------------------------|
| Disposal | : Seal the waste in vapour-tight plastic bags for eventual disposal by incineration. |
|----------|--------------------------------------------------------------------------------------|

### SECTION - 14 - TRANSPORT INFORMATION

|               | IATA         | IMO          | RID / ADR    |
|---------------|--------------|--------------|--------------|
| Shipping name | ISO PROPANOL | ISO PROPANOL | ISO PROPANOL |
| Hazard class  | 3            | 3            | 3            |
| UN Number     | 1219         | 1219         | 1219         |
| Packing Group | II           | II           | II           |



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DOT (UN) Transportation Information : Toxic-Inhalation Hazard.

|                |        |
|----------------|--------|
| UN Number      | : 1219 |
| Response Guide | : 129  |
| Hazard Class   | : 3    |
| Packing group  | : II   |

#### SECTION 15 - REGULATORY INFORMATION

Acute Low Toxic / Flammable substance

#### SECTION 16 - OTHER INFORMATION

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Prepared by                 | Deepak Fertilisers and Petrochemicals Corporation Ltd.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| Review No. & Date of Review | 3/1/2021                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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